PERMANENCE OF THE SYNTOMIC, SMOOTH AND ETALE PROPERTIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The three-in-one phrasing (syntomic, resp. smooth, resp. etale) of the statement and of the proof is the author's and is delivered verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Permanence of the syntomic, smooth and etale properties along a surjective such morphism. Morphisms, Lemmas morphisms-lemma-syntomic-permanence, morphisms-lemma-smooth-permanence and morphisms-lemma-etale-permanence-two, invoked at proof lines 4658-4661. Supply the conclusion once the finiteness input is available.
- Permanence of flat finite presentation along a flat surjective finitely presented morphism. Same chapter, Lemma lemma-flat-finitely-presented-permanence, invoked at proof line 4662. Gives the finiteness of q that the three permanence lemmas require.

Source reference. Original lemma 02KL: sources/stacks-descent/pinned-original-source.tex, source lines 4602–4602. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

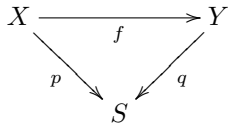
Source reference. Original lemma 02K6: sources/stacks-chow/tags.txt, source lines 3361–3361. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02K5: sources/stacks-chow/tags.txt, source lines 3360–3360. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02K3: sources/stacks-chow/tags.txt, source lines 3358–3358. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of schemes. Assume that

(1) *f is surjective, and syntomic (resp. smooth, resp. étale),*
(2) *p is syntomic (resp. smooth, resp. étale).*

Then q is syntomic (resp. smooth, resp. étale).

Proof. Combine Morphisms, Lemmas 02K3, 02K5, and 02K6 with Lemma 02KL above. □